
Demand shocks and firm-financed R&D expenditures

MATTHEW RAFFERTY and MARK FUNK*[‡]

School of Business, Quinnipiac University, Hamden, CT 06518, USA and

[‡]College of Business, University of Arkansas at Little Rock, Little Rock, AR 72104, USA

Business cycles might affect firms' ability and incentive to perform R&D. Firms finance most R&D activities out of cash flow so when cash flow decreases the funds available for R&D also decreases. This limits the ability of firms to perform R&D, potentially leading to reduced R&D expenditures during recessions. However, business cycles also influence the incentive to perform R&D. The opportunity cost of funds devoted to R&D falls during recessions since the return on production will likely be lower than during an expansion. During recessions, this provides firms with an incentive to redistribute an existing pool of funds away from production and towards R&D projects. This paper tests whether the business cycle influences the incentive and ability of firms to engage in R&D activities, in particular examining whether the response is symmetric across the business cycle.

I. INTRODUCTION

The business cycle might influence long run growth if the cycle influences productivity growth. Since firm-financed R&D expenditures are an important determinant of aggregate productivity, they provide a good place to start searching for a link between business cycles and long-run growth. The results in this paper suggest that an opportunity-cost effect exists for the largest R&D-performing firms. In addition, the effect of the business cycle on R&D expenditures is asymmetric. These links between the cycle and innovation-driven productivity growth have interesting implications for stabilization policy.

Theory cannot predict whether business cycles increase, decrease, or do not affect R&D expenditures. In models linking cash flow and R&D, cash flow acts as the pool of funds available to a firm. When the size of the pool increases, firms are able to engage in more R&D activities. This cash flow effect might be especially pronounced since

Hall (1992) points out that firms finance most R&D out of cash flow. Himmelberg and Petersen (1994) and Mulkay, Hall and Mairesse (2001) provide further empirical evidence of a significant positive relationship between R&D and cash flow.¹ Thus it might be expected that recessions reduce productivity growth since reduced cash flow leads to reduced R&D expenditures.

While the cash flow effect concerns the size of the pool of funds available to firms, how the firm allocates the pool over the firm's various activities may also respond to the business cycle. Bean (1990) and Saint-Paul (1993) argue that this allocation question means inventive activity should increase during recessions and decrease during expansions. In their view, firms engage in inter-temporal profit maximization with a tradeoff between current and future profits. Firms choose between increasing current profits by expanding output or increasing future profits by expanding inventive activity. Thus there is an opportunity cost to inventive activity: due to limited resources, a decision to increase current

*Corresponding author. E-mail: mffunk@ualr.edu

¹ The theoretical reasons for the existence of capital market imperfections in the financing of R&D are well known. For example, the nature of R&D makes it difficult for outsiders to predict the investment value, which prevents firms from borrowing to finance R&D projects in the same manner that a firm may borrow to build a new factory. The firm itself may seek secrecy for strategic reasons and may prefer internal financing of R&D. For a fuller explanation, see Hall (2002).

inventive activity is necessarily a decision to reduce resources devoted to current production. Moreover, since the production forgone during a recession results in relatively fewer forgone profits, an extra dollar devoted to inventive activity during a recession has a lower opportunity cost than it would during an expansion. As the opportunity cost of inventive activity falls during recessions one would expect that inventive activity increases. A similar argument suggests firms decrease inventive activity during expansions. Bean (1990) and Saint-Paul (1993) refer to this behaviour as the 'opportunity-cost' effect.

The opportunity-cost and cash flow effects both suggest a potential link between business cycles and long run growth, although they have opposing policy implications. The cash flow effect suggests that by reducing the long run growth rate recessions have more serious welfare consequences than conventional wisdom suggests. The opportunity-cost effect suggests the opposite: if R&D expenditures increase during recessions then recessions may stimulate long-run growth and increase long-run welfare.

Knowing whether both effects exist and whether the cash flow or opportunity-cost effect dominates is vital to understanding the effect of business cycles and demand shocks on long run growth. As Blackburn (1999) explains, if business cycles increase inventive activity then the current policy of smoothing out the business cycle might reduce productivity growth and long run growth. But if business cycles decrease inventive activity, the current policy might increase productivity growth and long-run growth.

Cash flow and opportunity-cost effects are tested for using firm-level data on R&D, sales, and cash flow, and an industry-level measure of aggregate demand. While the effect of cash flow on firm-level R&D has been studied, to the authors' knowledge no one has tested for the opportunity-cost effect using firm-level R&D data. Bean (1990), Gali and Hammour (1992) and Saint-Paul (1993) using national-level data found evidence suggesting positive aggregate demand shocks reduce the growth rate of total factor productivity (TFP). These findings suggest an opportunity-cost effect exists, but none of the studies were able to determine the mechanism through which positive demand shocks reduce TFP growth. Saint-Paul (1993) examined the R&D mechanism, but used national-level R&D and did not find evidence of an opportunity-cost effect in R&D investment. Since business cycles affect industries and firms differently, national-level data might average out demand shock effects on individual firms' R&D activities. This aggregation problem is avoided by using firm-level and industry-level data.

Additionally, the cash flow and opportunity-cost effects concern the ability and willingness of firms to engage in inventive activity. This means that a measure of inventive inputs to test for the effects is needed; TFP used in previous studies of the opportunity cost effect is a measure of inventive output. Firm-financed R&D is used as a measure of

inventive activity. Furthermore, national-level studies face data limitation problems; Saint-Paul (1993), for example, used only 20 years of data. Here, panel data is used, providing significantly more observations and improving the chances of estimating the cash flow and opportunity-cost effects, especially if the effects are relatively weak.

It might be argued that whether R&D is procyclical or countercyclical is of little interest since the effect on R&D of a recession is offset by the effect on R&D of the following expansion. However, the business cycle is not symmetric. Expansions are longer than recessions and the decrease in output during recessions might be steeper than the increase in output during expansions. Therefore, business cycle effects do not necessarily cancel out and the business cycle may either increase or decrease R&D. To account for this possibility, the opportunity-cost and cash flow effects are allowed to vary over the business cycle. By allowing for asymmetric effects, a better insight is gained into the relationship between cash flow, opportunity costs, and R&D. Even if the business cycle effects did cancel out, the knowledge that inventive activity responded to the business cycle could influence stabilization policy. If inventive activity is procyclical then policy makers would still want to eliminate recessions, but they might want to reconsider the policy of smoothing out expansions since that policy would reduce inventive activity.

Evidence is found suggesting R&D is generally procyclical, although not symmetric over the cycle. The cash flow effect is particularly strong during recessions, with only very weak effects during expansions. Moreover, the opportunity cost effect appears only with large firms, with some evidence to suggest it is a feature of expansions. The paper proceeds as follows. In Section II an error-correction model is developed that encompasses both the long run relationship between research and cash flow as well as the short run business cycle effects. The data is also described in Section II. The results of the panel error-correction approach are presented in Section III. Section IV provides conclusions and a discussion of the policy implications.

II. METHODOLOGY

Model

Following Mulkay *et al.* (2001), neoclassical investment theory is considered first, which implies that a desired level of the capital stock exists. A profit-maximizing firm with a constant elasticity of substitution (CES) production function using two inputs, the fixed capital stock (K) and the knowledge capital stock (G), with an elasticity of substitution parameter σ and returns to scale parameter ν , has the following long run optimal demand for knowledge

capital:

$$g_{it}^* = \alpha_{it} + \left(\sigma + \frac{1 - \sigma}{v} \right) s_{it} - \sigma c_{it} \quad (1)$$

where s_{it} is the natural log of output, g_{it} is natural log of the net knowledge capital stock, and c_{it} is the user cost of knowledge capital.² Mulkay *et al.* (2001) note the difficulties in measuring the user cost of capital for an individual firm. Their approach is to control for variations in the user cost of capital by using firm-specific and time-specific effects in estimation contained in the parameter α_{it} which takes the additive form $\alpha_{it} = \alpha_i + \alpha_t$. This paper follows their approach.

An autoregressive distributed lag (ADL) model is commonly used as the basis for examining the adjustment mechanism between the actual and desired capital stocks. An ADL(2,2) model of the form

$$g_{it} = \alpha_i + \alpha_t + \gamma_1 g_{i,t-1} + \gamma_2 g_{i,t-2} + \beta_0 s_{it} + \beta_1 s_{i,t-1} + \beta_2 s_{i,t-2} + \varepsilon_t \quad (2)$$

is used to start with. Since interest in both long run and short run changes in demand variables on the knowledge capital stock, it is useful to reparameterize Equation 2 into the following baseline error-correction model:³

$$\Delta g_{it} = \alpha_i + \alpha_t + \sum_{m=1}^1 \theta_j \Delta g_{i,t-j} + \sum_{l=0}^1 \varphi_l \Delta s_{i,t-l} + \rho(g_{i,t-1} - \lambda_1 s_{i,t-1}) + \varepsilon_{it} \quad (3)$$

The ECM framework encompasses both the long-run relationship between R&D and other variables and the short run effect of business cycles. In Equation 3 λ represents the cointegrating relationship between the knowledge capital stock and sales and ρ represents the speed of adjustment term.

Equation 3 has no direct role for either the cash flow or opportunity-cost effects. To test for the cash flow effect, Equation 3 is augmented with current and lagged values of cash flow. The equation becomes

$$\Delta g_{it} = \alpha_i + \alpha_t + \sum_{m=1}^1 \theta_j \Delta g_{i,t-j} + \sum_{l=0}^1 \varphi_l \Delta s_{i,t-l} + \sum_{l=0}^1 \eta_l \Delta cf_{i,t-l} + \rho(g_{i,t-1} - \lambda_1 s_{i,t-1} - \lambda_2 cf_{i,t-1}) + \varepsilon_{it} \quad (4)$$

where cf is after-tax cash flow. In this model, the short run effect of cash flow on the knowledge capital stock is given by the η 's and the short run effect of sales is given by the ϕ 's.

This baseline equation can be modified in several different ways to test for the presence of an opportunity-cost effect. In the first modification to Equation 4, an exogenous measure of demand for the firm's product, D_i or 'Demand' is included, defined as real shipments from the firm's (SIC 4-digit) industry. The advantage of this Demand variable over some variant of firm sales is that firm sales is a mixture of (endogenous) supply and demand forces while the 'Demand' variable is an exogenous measure of current and future demand for the firm's product. It is something firms observe, to a reasonable degree, and consider when determining R&D budgets.

Adding the Demand variable to Equation 4 creates

$$\Delta g_{it} = \alpha_i + \alpha_t + \sum_{m=1}^1 \theta_j \Delta g_{i,t-j} + \sum_{l=0}^1 \varphi_l \Delta s_{i,t-l} + \sum_{l=0}^1 \eta_l \Delta cf_{i,t-l} + \sum_{l=0}^1 \mu_l \Delta D_{i,t-l} + \rho(g_{i,t-1} - \lambda_1 s_{i,t-1} - \lambda_2 cf_{i,t-1}) + \varepsilon_{it} \quad (5)$$

where ΔD is the growth rate of Demand. If an opportunity-cost effect exists then an increase in demand will cause a temporary decrease in R&D investment, but over time R&D investment and the knowledge capital stock rise as per Equation 1. Therefore, it is expected that $\mu_0 < 0$, $\mu_1 < 0$ if the opportunity-cost effect plays a role in determining the knowledge capital stock. These are the key parameters for the opportunity-cost effect and are the focus of the remainder of the paper.

It might be expected that positive and negative shocks have different effects on the knowledge capital stock. For example, financing constraints tend to bind during recessions but not expansions. In addition, firms might respond to positive demand shocks by moving research workers to the shop floor to assist in production engineering or addressing specific customer product requirements. During negative demand shocks, firms might reduce their R&D spending or, given the high adjustment costs to R&D investment, choose to leave their research budget unchanged. If this is the case then the opportunity-cost effect might be a phenomenon of expansions, but not recessions.⁴

² This last term equals $c_{it} = (P_t^R/P_t)(r_t(P_{t-1}^R/P_t^I)) + \delta_t - (\Delta P_t^R/P_t^R)$ where δ_t is the depreciation rate, P_t^R is the price of a unit of knowledge capital, P_t^I is the price of a unit of physical capital, P_t is the price of industrial output at the sector level, and r_t is the real interest rate.

³ See Hendry *et al.* (1984) and Banerjee *et al.* (1993) for details.

⁴ A large portion of the R&D investment value resides in the brains of the research employees, creating a substantial adjustment cost to R&D investment. Firms may be reluctant to fire researchers, since this will cause the firm to lose a portion of the firm's overall knowledge stock. However, researchers moved temporarily to production will not cause the firm to lose the fruits of their previous R&D investment, but will slow the rate of knowledge growth. Thus, the adjustment cost will motivate firms to reallocate research resources within the firm. This behaviour should reduce the cash flow effect toward zero, while increasing the opportunity-cost effect.

To test for this possibility the growth rates of sales, cash flow and demand are interacted with a dummy variable, such that

$$\begin{aligned}\Delta g_{it} = & \alpha_i + \alpha_t + \sum_{m=1}^1 \theta_j \Delta g_{i,t-j} + \sum_{l=0}^1 \varphi_l \Delta s_{i,t-l} \\ & + \sum_{l=0}^1 \eta_l \Delta cf_{i,t-l} + \sum_{l=0}^1 \mu_l \Delta D_{i,t-l} \\ & + \xi Dummy_{it} \Delta s_{i,t} + \Psi Dummy_{it} \Delta cf_{i,t} \\ & + \zeta Dummy_{it} \Delta D_{i,t} + \rho(g_{i,t-1} - \lambda s_{i,t-1} - \lambda_2 cf_{i,t-1}) + \varepsilon_{it}\end{aligned}\quad (6)$$

where *Dummy* takes on the value of 1 if the variable's growth rate is negative and 0 otherwise. If the effects of shocks are symmetric then it is expected that $\xi = \Psi = \zeta = 0$.

Several methods exist for estimating Equations 5 and 6. Here, a two-stage procedure is used where the long run relationships, given by the λ 's, is estimated in the initial stage and then the error-correction models, Equations 5 and 6, are estimated in the second stage. While the primary focus is on the short run reactions to aggregate demand and cash flow shocks, reliable estimates of these reactions require consistent estimates of the long-run relationships. Of the several methods in the literature for estimating the long run relationships, the Pooled Mean Group (PMG, Pesaran *et al.*, 1999) maximum likelihood estimator comes closest to matching the needs here. The PMG estimator assumes a common long run relationship across firms, while allowing for potentially heterogeneous short run responses and error variances, and is appropriate whether the variables are $I(0)$ or $I(1)$.⁵ In practice, the PMG long-run coefficients are estimated jointly through a maximum likelihood procedure, followed by OLS estimation of the short run coefficients.⁶

Data

The firm-level data on research, cash flow, and sales come from the Compustat database. The industry shipments come from the NBER Manufacturing and Productivity Database which ends in 1996. Cash flow is only reported consistently from 1968, and since the sample is limited to full cycles the sample covers 1973–1990. The sample is limited to manufacturing firms (SIC 2 or 3). Firms are included only if they conducted research over the entire

period. Firms incorporated in foreign countries and wholly-owned subsidiaries of foreign firms were deleted. These considerations limited the sample to 94 manufacturing firms during 1973–1990 which provides 1598 observations.

Sales and after-tax cash flow are deflated by SIC 4-digit industry-specific shipment deflators taken from the Manufacturing Industry Productivity Database at NBER. Since firms expense rather than capitalize research, research expenses are added back into reported cash flow to reflect the cash available for research and other activities. Research spending is deflated by the implicit research deflator in the NSF's *Science and Technology Indicators*. Following convention, the research stock is computed by the perpetual inventory method with a constant rate of depreciation set at 15% for all firms. Table 1 provides some sample characteristics.

III. EMPIRICAL RESULTS

Table 2 contains the results of estimating Equation 5 using several different sample sizes and specifications.⁷ The long run relationship estimated by the PMG method and used in Equations 5 and 6 is $0.683 * s + 0.454 * cf$, with both coefficients significant at the 95% level. The results are consistent with previous work examining the relationship between R&D and cash flow – increased cash flow stimulates R&D. However, there is little evidence supporting the opportunity–cost effect. The coefficient on the demand variable is negative, but is not statistically significant at either the 95% or 90% confidence level.

The first column of results in Table 2 assumes identical cash flow and opportunity-cost effects for large and small firms. It is already common to examine the influence of firm size on the cash flow effect (e.g., Himmelberg and Peterson, 1994) and it is probable that the opportunity-cost effect also varies with firm size. The degree of substitutability between inventive activity and current production may increase with the research budget, since firms with a greater variety of research activities enjoy more substitution possibilities. Firms with more research activities are more likely to have research projects of varying priority, including low priority projects they may be willing to shelve to free resources for more production to take advantage of temporarily high demand. Firms may prioritize research projects for a variety of reasons: differences in the ability

⁵ Im *et al.* (1997) panel unit root tests indicate our data contain a unit root. Formal panel cointegration tests (Kao, 1999) confirm the cointegrating relationship indicated by the significant negative ECM term.

⁶ While we use Pesaran's *et al.* (1999) maximum likelihood procedure to estimate the long run relationship, alternative long run estimators could be employed in similar fashion. Experiments with alternative estimators ranging from individual OLS to panel cointegration estimates returned similar long run relationships, which were similar to those reported previously for sales and cash flow (e.g., Mulkay *et al.*, 2001). The short run reactions—which are our primary interest—proved robust to the method chosen to estimate the long run relationship.

⁷ All results include fixed effects and time period dummies which are available upon request.

Table 1. *Data characteristics US\$ in millions*

	1990 average	1990 median	1973 average	1973 median
Full sample				
R&D expenditures	74.30	10.40	32.79	3.82
Cash flow	73.85	13.27	44.10	5.76
R&D stock	364.23	57.82	180.53	18.51
Sales	2899.93	487.87	2256.09	267.28
Industry shipments	22 333.41	14 390.22	16 873.77	10 490.79
Small research firms				
R&D expenditures	6.87	3.45	1.57	1.15
Cash flow	14.18	4.66	3.21	1.97
R&D stock	30.40	13.97	7.39	6.00
Sales	285.31	150.35	137.60	79.08
Industry shipments	14 838.17	9777.04	9823.79	4753.40
Large research firms				
R&D expenditures	141.75	34.97	64.01	20.44
Cash flow	133.53	51.63	85.00	18.30
R&D stock	698.06	199.34	353.67	118.25
Sales	5514.55	2079.03	4374.57	1037.52
Industry shipments	29 828.65	17 156.04	23 923.75	14 983.42

Note: Number of firms: 94.

Table 2. *Estimated error-correction models*

Variable	Full sample Number of firms: 94 number of observations: 1598	Large firms Number of firms: 47 Number of observations: 799	Small firms Number of firms: 47 Number of observations: 799
Dependent variable: R&D stock growth rate			
Estimation method: Fixed effects			
Period: 1974–1990			
Sales growth gate	0.080 (0.015) [0.000]	0.074 (0.008) [0.000]	0.086 (0.025) [0.001]
Sales growth rate (−1)	0.018 (0.015) [0.247]	0.017 (0.008) [0.037]	0.023 (0.022) [0.286]
Cash flow growth rate	0.031 (0.005) [0.000]	0.016 (0.004) [0.000]	0.037 (0.007) [0.000]
Cash flow growth rate (−1)	0.006 (0.004) [0.125]	−0.000 (0.002) [0.996]	0.009 (0.005) [0.070]
Demand growth rate	−0.017 (0.014) [0.220]	−0.017 (0.010) [0.109]	−0.007 (0.026) [0.798]
Demand growth rate (−1)	0.006 (0.016) [0.685]	0.007 (0.010) [0.438]	0.012 (0.025) [0.614]
R&D stock growth rate (−1)	0.445 (0.069) [0.000]	0.615 (0.036) [0.000]	0.425 (0.073) [0.000]
Error-correction term (−1)	−0.062 (0.020) [0.002]	−0.028 (0.004) [0.000]	−0.071 (0.026) [0.006]

Note: Coefficient estimate appears first in each cell. Newey–West standard errors are in parentheses. *P*-values are in brackets.

Table 3. *Estimated error-correction models (full sample with asymmetric effects)*

Variable	Full sample Number of firms: 94 Number of observations: 1598	Large firms Number of firms: 47 Number of observations: 799	Small firms Number of firms: 47 Number of observations: 799
Dependent variable: R&D stock growth rate			
Estimation method: Fixed effects			
Period: 1974–1990			
Sales growth rate	0.105 (0.015) [0.000]	0.076 (0.015) [0.000]	0.122 (0.021) [0.000]
Sales growth rate (–1)	0.016 (0.015) [0.279]	0.017 (0.008) [0.030]	0.021 (0.022) [0.345]
Cash Flow Growth Rate	0.014 (0.005) [0.005]	0.008 (0.005) [0.124]	0.016 (0.007) [0.021]
Cash flow growth rate (–1)	0.007 (0.004) [0.070]	0.001 (0.003) [0.793]	0.011 (0.005) [0.037]
Demand growth rate	–0.021 (0.024) [0.400]	–0.035 (0.017) [0.035]	0.008 (0.047) [0.863]
Demand growth rate (–1)	0.014 (0.014) [0.346]	0.010 (0.009) [0.261]	0.019 (0.025) [0.430]
Dummy * Sales growth rate	–0.053 (0.031) [0.086]	–0.007 (0.020) [0.750]	–0.083 (0.062) [0.183]
Dummy * Cash flow growth rate	0.041 (0.013) [0.001]	0.021 (0.009) [0.013]	0.052 (0.019) [0.006]
Dummy * Demand growth rate	0.008 (0.057) [0.894]	0.032 (0.027) [0.237]	–0.032 (0.135) [0.811]
R&D stock growth rate (–1)	0.427 (0.068) [0.000]	0.596 (0.037) [0.000]	0.419 (0.071) [0.000]
Error-correction term (–1)	–0.064 (0.020) [0.001]	–0.030 (0.004) [0.000]	–0.074 (0.026) [0.005]
Wald test for asymmetric effects of business cycle	11.619 [0.009]	7.465 [0.058]	8.841 [0.031]

Note: Coefficient estimate appears first in each cell. Newey–West standard errors are in parentheses. *P*-values are in brackets. For the Wald tests, the test statistic appears first with *p*-value in brackets. Each test statistic has three degrees of freedom.

to appropriate returns to basic versus developmental research; lower expected returns for some projects; projects in areas outside the firm's core business etc. If some projects are more critical than others then firms may be willing to alter the pace of low priority projects in response to the business cycle.

To test the joint hypothesis that the opportunity-cost effect exists and substitution possibilities increase with the amount of R&D expenditures, the firms are divided into 'large' and 'small' firm subsamples, with 'large' firms defined as the firms above the full sample median R&D expenditures for 1973. The 'large firms' results in Table 2 provide more support for the opportunity-cost effect. The coefficients on

the current growth rate of Demand are negative and just statistically significant at the 90% level. The effect seems strong since a 1% increase in Demand is associated with a nearly 2% decrease in the short run knowledge capital stock. This suggests that if an opportunity-cost effect exists then the size and range of a firm's activities play an important role in generating the effect. The cash flow effect is also weaker, consistent with previous findings that the research budgets of large firms are not as cash sensitive. The results for small firms shown in the third column of Table 2 provide little evidence of an opportunity-cost effect, consistent with the view that large research firms have more substitution possibilities than do small firms.

Table 3 contains the results of Equation 6 allowing for asymmetric effects for the full sample. Once again, the results for the full sample provide little evidence for the opportunity-cost effect; the coefficient on the current growth rate of demand is negative but statistically insignificant, and there is no indication of an asymmetric response. However, the interaction between cash flow growth and the dummy variable is statistically significant and positive. This suggests that the already significant cash flow effect is much stronger when cash flow is declining than when cash flow is increasing. Based upon these estimates, one would expect that a 1% decrease in cash flow would cause approximately a 6% decrease in the knowledge capital stock.

The second column of Table 3 shows the asymmetric results for the largest firms. The coefficient on the demand growth rate is negative and statistically significant at the 95% level, providing considerably more support for the opportunity-cost effect than any of the other models. The coefficient on the interaction between demand growth and the dummy variable is positive although statistically insignificant. Since the magnitude is roughly the same as the coefficient on demand growth by itself, the interaction coefficient suggests that firms may not respond to negative demand shocks, but may decrease R&D in response to positive demand shocks. As for cash flow, the large firms respond primarily to negative shocks: while the cash flow coefficient is insignificant at the 90% level, the coefficient on the interaction term between cash flow growth and the dummy is nearly three times as large and significant at the 98% level.

The third column of Table 3 shows the results for the smallest firms. Again, the small firm results provide little indication of any opportunity-cost effect, while strongly suggesting a cash flow effect. In addition, the cash flow effect is again much stronger in response to decreases in cash flow than increases in cash flow.

Table 3 also shows the results of Wald tests for asymmetric effects of the business cycle. If the business cycle has symmetric effects then it would be expected that the coefficient on all three interaction terms would equal zero for all samples. However, the results reject the null-hypothesis of symmetric effects at the 95% level for the full sample and the smallest firms and the results reject the null at the 94% level for the largest firms. These results strongly suggest that the effect of the business cycle is asymmetric.

IV. CONCLUSION AND POLICY IMPLICATIONS

Firm-level R&D provides one possible channel linking business cycles and growth. This paper finds evidence suggesting that while R&D is generally procyclical, the response to the cycle is not symmetric. In particular, the

response to negative cash flow movements is much stronger than the response to positive cash flow movements, for both large and small firms. The opportunity-cost effect only appears for large research firms, possibly because firms with large budgets have more substitution possibilities between research and production. However, some evidence suggests the opportunity-cost effect only appears during expansions, as firms shift research input into production.

These findings support the current policy of smoothing away recessions. The strong response to negative cash flow shocks, combined with the absence of an opportunity-cost effect, implies that the potential welfare costs of recessions are greater than conventionally thought: by reducing research, recessions may reduce long-run productivity growth. Whether expansions should also be smoothed out is less clear. Small firms clearly benefit from cash flow increases and show no signs of an opportunity-cost effect, but large firms have only a weak cash flow response and appear to respond to positive demand shocks by shifting research resources into production via the opportunity-cost effect. Smoothing out expansions may increase research in large firms, at the cost of reducing research in small firms.

Using firm-level data, this paper finds results indicating that an important inventive activity, firm-financed R&D, responds to the business cycle. Since firm-financed R&D expenditures are an important determinant of productivity growth, the results also suggest that business cycles influence productivity growth. Given the importance of productivity growth in determining welfare in the long run, policy makers need to consider these effects when determining the appropriate stabilization policy.

REFERENCES

- Banerjee, A., Dolado, J., Galbraith, J. and Hendry, D. (1993) *Co-Integration, Error-Correction, and the Econometric Analysis of Non-Stationary Data*, Oxford University Press, New York.
- Bean, C. R. (1990) Endogenous growth and the procyclical behaviour of productivity, *European Economic Review*, **34**, 355–63.
- Blackburn, K. (1999) Can stabilization policy reduce long run growth?, *Economic Journal*, **109**, 67–77.
- Gali, J. and Hammour, M. L. (1992) *Long Run Effects of Business Cycles*, Columbia First Boston Series in Money, Economics, and Finance Working Paper: FB-92-26.
- Hall, B. (1992) *Investment and Research and Development at the Firm Level: Does the Source of Financing Matter?* Working Paper 4096, National Bureau of Economic Research, Cambridge, MA.
- Hall, B. (2002) The financing of research and development, *Oxford Review of Economic Policy*, **18**(1), 35–51.
- Hendry, D. F., Pagan, A. R. and Sargan, J. D. (1984) Dynamic specification, in *Handbook of Econometrics*, Vol. II (Eds) Z. Griliches and M. Intriligator, North-Holland, New York, pp. 1025–100.

- Himmelberg, C. and Petersen, B. (1994) R&D and internal finance: a panel study of small firms in high-tech industries, *Review of Economics and Statistics*, **76**(1), 38–51.
- Im, K., Pesaran, M. H. and Shin, Y. (1997) *Testing for Unit Roots in Heterogeneous Panels*, Department of Applied Economics Working Paper, University of Cambridge, revised.
- Kao, C. (1999) Spurious regression and residual-based tests for cointegration in panel data, *Journal of Econometrics*, **90**, 1–44.
- Mulkay, B., Hall, B. and Mairesse, J. (2000) Investment and R&D in France and the United States, NBER Working Paper, No. 8038.
- Pesaran, M. H., Shin, Y. and Smith, R. (1999) Pooled mean group estimation of dynamic heterogeneous panels, *Journal of the American Statistical Association*, **94**, 621–34.
- Saint-Paul, G. (1993) Productivity growth and the structure of the business cycle, *European Economic Review*, **37**, 861–90.

Copyright of Applied Economics is the property of Routledge, Ltd. and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.

Copyright of Applied Economics is the property of Routledge and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.